Detecting AI-Generated Text Using Lightweight Linguistic Models

Tyler Geiger and Kadin Matotek
Instructor: Dr. Agah
West Chester University, Department of Computer Science

Introduction

As LLMs like ChatGPT become more capable, it's getting harder to tell AI-generated text apart from human writing. This matters for academic integrity, journalism, and trust online.

We ask: **can a simple, interpretable model do the job?** Instead of a complex neural detector, we use logistic regression with TF-IDF features.

Research Questions:

- Can lightweight models reliably detect AI-generated text?
- Which linguistic patterns most distinguish ChatGPT from humans?
- Does detection hold across different topic domains?

Accuracy by Source

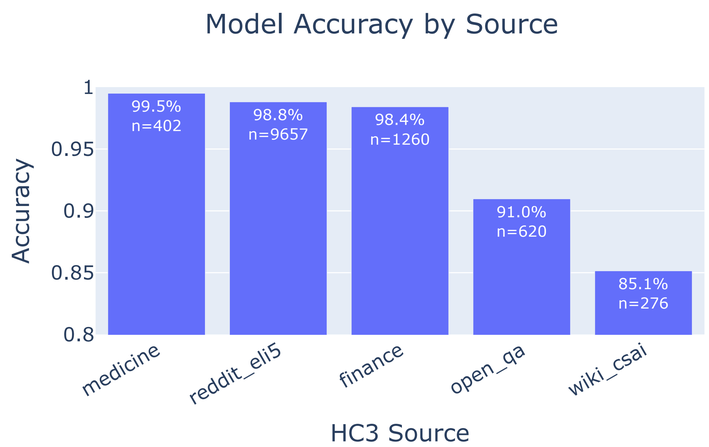

Figure 1. All five HC3 domains exceed 85% accuracy, indicating the model learns **authorship style**, not domain-specific vocabulary.

Methodology

Dataset (HC3):

- Human vs. ChatGPT answers to the same questions
- Domains: Reddit (ELI5), open QA, Wikipedia (CS), medicine, finance
- Split by question group to prevent leakage

Model:

- TF-IDF (50k features, unigrams + bigrams)
- Logistic Regression (balanced classes)
- Train/val/test split: 75% / 10% / 15%

Model Confidence Distribution

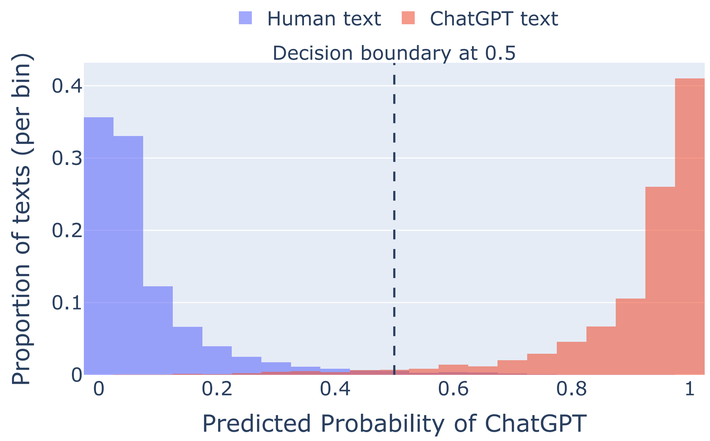

Figure 2. 97.4% of ChatGPT texts are correctly classified; 1.6% of human texts are misclassified. Human texts (blue) cluster near 0 and ChatGPT texts (orange) near 1, with a clear separation at 0.5. The model makes **high-confidence predictions**, reflecting uniform AI style and varied human writing.

Feature Importance

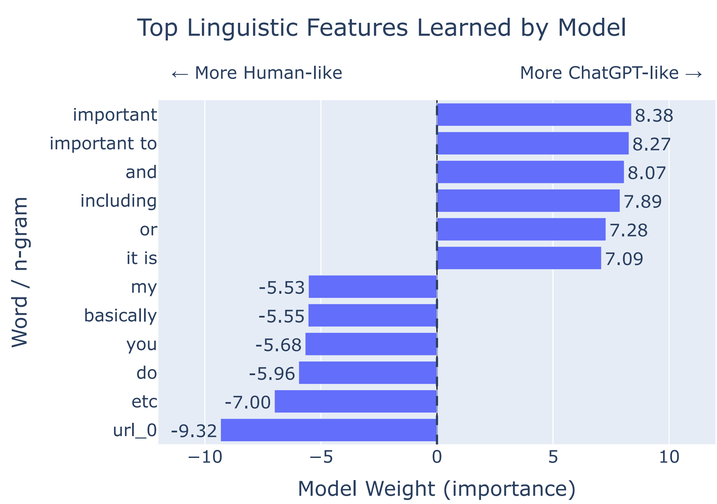

Figure 3. ChatGPT-indicative features (right) include common, generic terms such as *"important," "including,"* and *"it is."* Human-indicative features (left) include more informal and personal language like *"my," "you,"* and *"basically."* This contrast helps explain why a linear model separates the two effectively.

Findings

- >95% **accuracy** on held-out test data
- ChatGPT text is **stylistically uniform**, while human text is more diverse
- ChatGPT features skew toward **common, generic terms**, while human features reflect more personal and informal language
- Performance generalizes across all five domains

Takeaways:

- Lightweight models achieve strong performance at a fraction of the cost of neural detectors
- Interpretability allows us to **explain model decisions**, which is valuable in real-world settings
- As LLMs improve, lexical signals may weaken, requiring more advanced stylistic features

 km998744@wcupa.edu, tg996676@wcupa.edu